Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/18/2026
PI Name:	Stankovic, Aleksandra Ph.D.		
Project Title:	Quantification of Response to Virtual Reality-based Sensory Stimulation for Relaxation and Therapeutic Release in ICE		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	astankovic1@mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	908-391-1177
Organization Name:	Massachusetts General Hospital/Harvard Medical School		
PI Address 1:	Human Performance Laboratory		
PI Address 2:	149 13th St		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	09/09/2020	**End Date:**	09/08/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/08/2025 per L. Juliette/HFBP (Ed., 6/4/24). NOTE: End date changed to 09/08/2024 per L. Juliette/HFBP (Ed., 8/15/23).		
Key Personnel Changes/Previous PI:	Per the PI, the Co-Investigator during the final phase of the project was Jay Buckey, Ph.D.		
COI Name (Institution):	Buckey, Jay M.D. (Dartmouth College)		
Grant/Contract No.:	80NSSC20K1852		
Performance Goal No.:			
Performance Goal Text:			

	Long-duration space missions expose astronauts to prolonged isolation, confinement, and sensory monotony, which can negatively affect mood, cognition, and performance. This project demonstrated that virtual reality can provide an effective and well-accepted means of delivering restorative sensory experiences in spaceflight-like environments. Across laboratory and Antarctic analog studies, participants consistently reported improved mood, reduced stress, and enhanced sense of restoration following VR exposure. The first phase of this investigation involved the analysis of subjective feedback questionnaires and post-mission interviews collected from participating operational ICE environment volunteers who interacted with a standard VR platform on an informal basis. The purpose of this exploratory, opportunistic research was to assess preference for VR scenarios and to gather contextually-specific experiential data with the goal of optimizing future VR presentation for maximum restorative impact. This work has shown immersive VR to be highly rated, with natural scene content and dynamic scenes involving people and animals perceived as restorative following long periods of isolation and confinement. Findings suggest that options for personalized customization of the VR experience are also highly desirable.
Task Description:	We deployed a set of VR experiences for usage at the South Pole Station, and have collected data during the past two winter-over seasons. These experiences included several different VR scenarios which modulated one or more of four specific attributes of the VR experience: (1) scenario duration (short vs long); (2) sensory modality (visual only or visual haptic cues and enhanced audio); (3) scene context (city or nature scenes); and (4) scene dynamic presentation (fixed scenes or dynamically explorable scenes with motion). We also investigated differential effects of assigned vs. self-selected VR sessions. For both seasons, we collected subjective response data through self-reported mood and preference questionnaires, as well as objective physiological responses to VR experience interactions during the second half of winter-over (approximately July-October), to assess the emotional and psychological impacts of various platform configurations. Finally, a laboratory study was conducted to expand upon the analog work and incorporate an investigation of the impact of various VR experiences (including the introduction of biofeedback) on cognitive performance. The research showed that natural environments, longer experiences, and immersive presentation tend to produce the strongest benefits, while individual preferences vary considerably. Importantly, giving users autonomy in selecting and scheduling VR experiences enhanced acceptability and outcomes. These findings support the development of flexible, personalized VR countermeasures to promote behavioral health and performance during future exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Virtual Reality platforms offer tremendous promise, both on Earth and in spaceflight, as psychological support tools in conditions of prolonged isolation and confinement.
	This project aimed to optimize and test Virtual Reality (VR) sensory presentation for behavioral health support in isolated, confined, and extreme (ICE) environments. Across the funding period, the project successfully completed all three specific aims. In Aim 1, VR systems were deployed in operational ICE environments on an informal, voluntary basis. Subjective assessments demonstrated high acceptance, perceived restoration, and emotional impact, particularly for immersive natural environments and dynamic scenes involving animals or human activity. These findings demonstrated the viability and acceptability of VR in these environments, established feasibility, and informed subsequent experimental design (see Anderson & Stankovic et al., 2023). Aim 2 involved two consecutive winter-over deployments at the South Pole Station. Participants were exposed to VR conditions that manipulated scene content (natural versus urban), duration, sensory augmentation, and level of interactivity. Subjective mood, perceived restoration, and immersion ratings were collected alongside objective psychophysiological measures, including heart rate variability and electrodermal activity. Results demonstrated overall preferences for VR compared to non-immersive media, longer exposure durations, and nature-based content, while also revealing meaningful individual differences in scene preference and sensory tolerance (see Thoolen et al., 2025). Feedback from the first season directly informed refinements implemented in the second season, including greater user autonomy in scheduling and scene selection. This adaptive approach resulted in improved acceptability and supported the conclusion that personalization is a critical factor for optimizing VR-based countermeasures in ICE contexts. Aim 3 laboratory studies further demonstrated that VR exposure following stress induction supported psychophysiological recovery and attention restoration, with biofeedback-responsive VR showing promise for enhancing engagement. A supplemental parabolic flight study examined neurovestibular responses to VR under altered gravity conditions and found no significant adverse effects, supporting the feasibility of VR use during spaceflight. Methods: The first phase of this project involved analyzing subjective feedback questionnaires and post-mission interviews collected from participating members of multiple ICE environment pilot cohorts (e.g., HI-SEAS, CPS Alert) who interacted informally with a standard VR platform. In this exploratory and opportunistic research, immersive VR was highly rated, with natural scene content and dynamic scenes involving people and animals perceived as restorative. Personalized customization of the VR experience was rated as highly desirable (Anderson & Stankovic et al., 2023). To determine which VR attributes are most beneficial for improving mood including relaxation, and providing restoration in a high-fidelity long-duration ICE environment, we collected data from participating crewmembers at Amundsen-Scott South Pole Station in Antarctica during winter-over seasons 2022 and 2023. Findings from our first field season were published in npj Microgravity (Thoolen et al., 2025) and are detailed below. Via laboratory study, we investigated the impact of various aspects of VR (including the presentation of biofeedback cues) on mood, relaxation, restoration, and cognitive functioning. Stress and mood were measured through psychophysiology and self-report, and cognitive performance/attention restoration were assessed via an established, brief variant of the psychomotor vigilance task (PVT). As an opportunistic extension to the main Antarctic and laboratory studies, the project conducted a pilot investigation of virtual reality use during parabolic flight to evaluate neurovestibular responses to immersive sensory stimulation in altered gravity. This addendum was designed to address a key operational unknown: while VR systems had shown promise for behavioral health and performance support on Earth and in isolated analog environments, little empirical

Task Progress:	work had examined how VR exposure might interact with changing gravitational states. The study therefore, aimed to assess the feasibility and safety of VR use in microgravity and to generate preliminary design guidance for future spaceflight applications. Two healthy adult participants completed both ground-based and parabolic flight sessions, viewing a set of short VR scenes that varied by gravity context (1G or 0G) and motion cues (stationary or moving viewpoints). During the flight, participants experienced 13 microgravity parabolas while wearing a wireless VR headset and a physiological monitoring wristband that recorded heart rate, electrodermal activity, skin temperature, and blood volume pulse. After each scene, participants reported perceived immersion, disorientation, and nausea using short subjective rating scales. The pilot results demonstrated the feasibility of collecting both subjective and psychophysiological data during VR exposure in parabolic flight and provided an initial proof of concept for studying immersive systems in changing gravity environments (Harris et al., 2022). Although the small sample size and equipment limitations prevented formal statistical analysis, subjective reports indicated no nausea across any scenes and only minimal disorientation, while perceived immersion tended to increase when the VR content was matched to the experienced gravity condition. Physiological trends were variable and inconclusive, reflecting the exploratory nature of the study and the short duration of microgravity exposures, but no major adverse responses were observed. The work therefore established an initial methodological framework for evaluating VR use in microgravity and highlighted key considerations for future research, including improved physiological instrumentation, more sensitive subjective measures, and longer-duration testing. Overall, the parabolic flight study provided early evidence that immersive VR can be safely tolerated during altered gravity and offered practical insights into how scene design and gravity congruence may influence user experience, supporting continued development of VR-based behavioral health and operational tools for spaceflight. Results and Conclusions: Across pilot deployments, Antarctic field studies, laboratory experiments, and parabolic flight investigations, this project demonstrated that immersive virtual reality is a feasible, well-accepted, and effective behavioral health countermeasure in isolated, confined, and extreme environments. Early pilot work in analog sites confirmed strong user acceptance and high ratings of presence, emotional impact, and perceived restoration, with naturalistic scenes and dynamic content consistently rated as the most beneficial. These findings informed the design of controlled studies at the Amundsen-Scott South Pole Station, where winter-over crews completed structured exposure sessions under multiple sensory and presentation conditions. Across these studies, participants showed clear preferences for immersive VR over non-immersive alternatives, for nature-based scenes over urban content, and for longer exposure durations. VR sessions were associated with improvements in mood, perceived restoration, and immersion, consistent with greater relaxation and engagement. The Antarctic experiments also revealed important individual differences in content preferences and sensory tolerance. While nature scenes produced the most consistently positive mood outcomes, some participants reported relaxation benefits from urban environments, and reactions to additional sensory augmentation were mixed. These findings highlighted the importance of personalization, autonomy in scheduling, and flexible content selection. Subsequent study phases explored the extent to which providing participants with the ability to choose scene type, timing, and duration improved overall experience ratings and mood outcomes. The complementary laboratory study characterized how VR exposure following stress induction supported cognitive performance and emotional recovery, and an opportunistic parabolic flight experiments suggested that viewing Earth-referenced VR content in altered gravity conditions did not produce significant vestibular disturbances, supporting operational feasibility in spaceflight contexts. Collectively, the results established VR as a viable and adaptable tool for promoting relaxation, mood regulation, and cognitive resilience in long-duration missions. Impact: The project provides a strong empirical foundation for the use of immersive virtual reality as an operational behavioral health countermeasure for future lunar, cis-lunar, and Mars missions. By demonstrating feasibility, acceptability, and measurable mood and restoration benefits across multiple isolated and extreme environments, the work shows that VR can help mitigate the sensory deprivation, monotony, and psychological stressors inherent to long-duration spaceflight. The findings indicate that the most effective systems will not rely on a single standardized experience, but instead will provide adaptable, personalized content and allow crew autonomy in selecting scenes, duration, and scheduling. This approach aligns with the broader shift toward crew-autonomous behavioral health support, where astronauts can independently manage stress, attention, and mood without continuous ground intervention. The project's results also inform design requirements for future VR systems, including content selection strategies, exposure duration guidelines, and integration with physiological monitoring and biofeedback, as well as operational implications for crew time scheduling with respect to engagement with behavioral health and performance countermeasures. Together, these insights support the development of optimized, flight-ready VR platforms capable of sustaining cognitive performance, emotional well-being, and mission readiness during extended exploration missions.
Bibliography Type:	Description: (Last Updated: 03/18/2026)
Abstracts for Journals and Proceedings	Harris K, Hee R, Hoffman J, Stankovic A. "Neurovestibular response to Virtual Reality sensory presentation in mixed-gravitational conditions." 73rd International Astronautical Congress, Paris, France, September 18-22, 2022. Abstracts. 73rd International Astronautical Congress, Paris, France, September 18-22, 2022. , Sep-2022
Articles in Peer-reviewed Journals	Anderson A, Stankovic A, Cowen D, Fellows A, Buckey JC. "Natural scene virtual reality as a behavioral health countermeasure in isolated, confined, and extreme environments: Three isolated, confined, extreme analog case studies." Hum Factors. 2023 Sep;65(6):1266-1278. https://doi.org/10.1177/00187208221100693 ; PubMed PMID: 35604867 , Sep-2023
Articles in Peer-reviewed Journals	Thoolen SJJ, Watson JM, Fellows AM, Bovard PP, Strangman GE, Buckey JC, Stankovic AS. "Virtual reality-based sensory stimulation preferences at Amundsen-Scott South Pole Station in Antarctica." npj Microgravity. 2025 May 11;11(1):15. https://doi.org/10.1038/s41526-025-00471-2 ; PMID: 40350468 PMCID: PMC12066713 , May-2025